BMS CHO Cell Productivity Knowledge Graph

User Manual — Version 3

Elucidata × BMS

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1 Welcome

1.1 About this Manual

This manual is your end-to-end guide for working with the **BMS CHO Cell Productivity Knowledge Graph (KG)**, a multi-omics, multi-evidence platform built to accelerate the identification of genetic engineering targets that improve CHO cell productivity.

The KG integrates:

- **BMS in-house data** — transcriptomics, proteomics, and metabolomics from Study 1 and Study 2 alongside bioprocess readouts (qP, titre, IVCD, lactate, cell density)
- **CHO-specific public omics data** — 13 datasets from 10+ publications comparing high vs. low producers
- **Causal literature evidence** — cause-effect relationships curated from ~290 genetic perturbation publications
- **External databases** — NCBI, UniProt, GO, KEGG, ChEBI, PubMed, Cellosaurus

The bioinformatics analyses embedded in the KG include differential expression / abundance (DEG/DAA), WGCNA co-expression networks, MOFA multi-omics integration, pathway enrichment (GSEA & ORA), and transcription-factor enrichment.

1.2 How to Use this Manual

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1.3 Manual Structure

This manual is organised into nine chapters, followed by a glossary of key terms.

1.4 Key Conventions

> **Note**
>
> **Query boxes** throughout this manual show verbatim natural-language queries you can type directly into the KG interface. Results may vary slightly depending on cohort or filter settings.

> **Tip**
>
> **Evidence badges** tag information by analytical layer:
> DEG/DAA WGCNA MOFA Literature GEM (in progress)

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2 Getting Started

How to access the KG, navigate the interface, and run your first query

2.1 Overview

This chapter walks you through logging into the BMS KG platform (Polly Co-Scientist), understanding the workspace layout, and running your first natural-language query. It takes about five minutes.

2.2 Accessing the Platform

The BMS Knowledge Graph is hosted on **Polly** (Elucidata's life-sciences data platform) and accessed through a browser — no installation required.

Login steps:

> **Note**
>
> If you cannot log in or do not see the BMS KG workspace, contact your Elucidata administrator or email yogesh.lakhotia@elucidata.io.

2.3 The Interface at a Glance

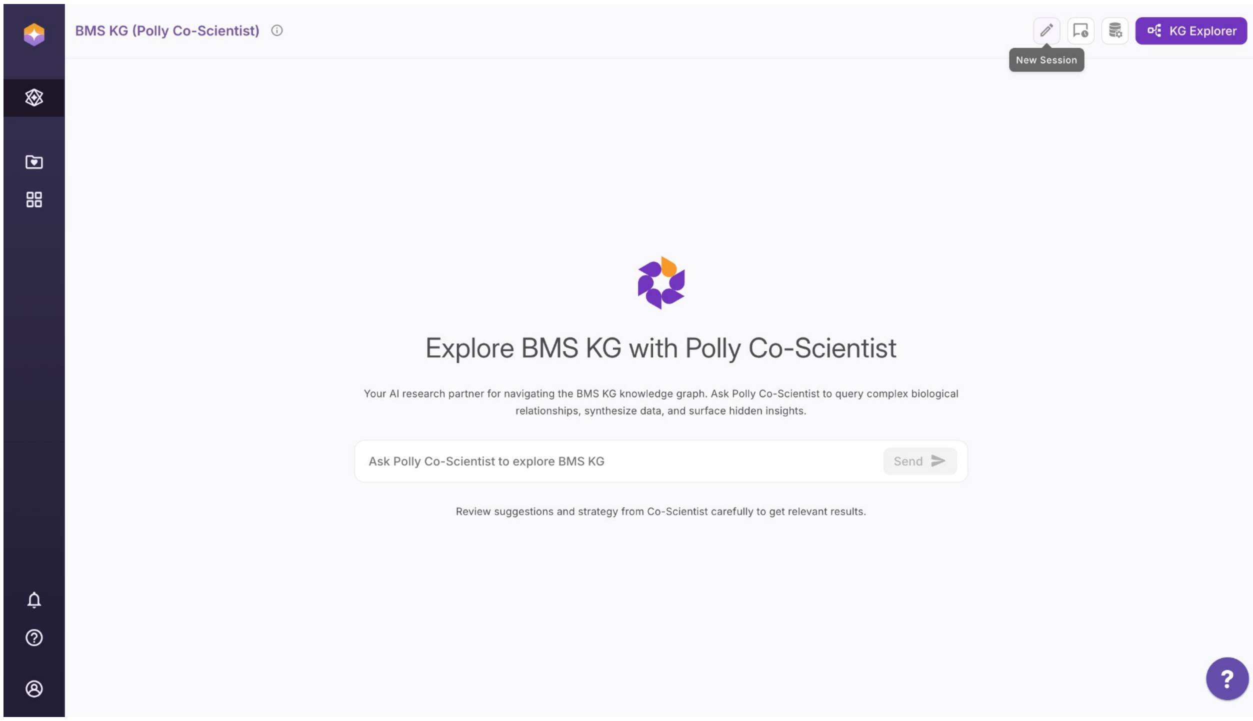

Figure 2.1: The BMS KG interface — Polly Co-Scientist. Key controls are highlighted: New Session (pencil icon), Chat History (clock icon), KG Explorer (top right), and the central query bar.

The interface has four main areas:

Area	Location	What it does
Left sidebar	Far left icon strip	Navigation — workspace, files, apps, settings
Query bar	Centre of screen	Type your natural-language question here and press Send
New Session	Top right (pencil icon)	Starts a fresh conversation with no prior context
Chat History	Top right (clock icon)	Browse and resume previous sessions
KG Explorer	Top right (purple button)	Switch to visual graph exploration mode

2.4 Starting a New Session

Each session maintains its own conversation context. Start a new session when:

- You are switching to a completely different question or molecule
- A previous session has accumulated too much context and responses feel off-track
- You want a clean slate for a structured analysis workflow

2.5 Viewing Chat History

All sessions are automatically saved.

2.6 KG Explorer Mode

KG Explorer is a **visual graph browser** — it lets you navigate the knowledge graph as an interactive node-edge network rather than through conversation.

> Tip
>
> Use KG Explorer when you want to **visually trace** relationships from a specific gene — for example, following HSPA5 through its WGCNA module, to its enriched pathways, to its TF regulators, all in one graph view.

2.7 Exporting a Chat Session

To share your analysis or save results:

2.8 Your First Query

Once logged in, try this to verify the KG is working and to get an immediate sense of what is available:

What molecules and clones are available in the BMS internal dataset?

A working KG will return a table listing Nivo and C144 clones with their study and productivity group classifications. If you see an error or an empty result, contact your administrator.

2.9 Tips Before You Start

- **Be specific about the entity type** — say "gene", "protein", or "metabolite" explicitly
- **Name the cohort or condition** — include molecule (Nivo / C144) and study when relevant

- **Include thresholds** — e.g., "adj. p-value < 0.05 and $|\text{log2FC}| > 2$"
- **Review suggestions carefully** — Co-Scientist may offer a query strategy before running; check it makes sense before confirming

Full query-writing guidance is in Section 3.6.

3 Orientation

KG structure, data model, cohort naming, and how to phrase queries

3.1 Overview

Before running any analysis, spend a few minutes here. Users who understand the KG structure get significantly better query results and spend less time troubleshooting empty returns.

3.2 What is the Knowledge Graph?

The BMS CHO Productivity Knowledge Graph is a **property graph database** (Neo4j) that connects biological entities — genes, proteins, metabolites, pathways, samples, publications — through typed, directed relationships. It is the integration layer that lets a single question simultaneously draw on multi-omics data, co-expression networks, MOFA factors, pathway annotations, and published literature evidence.

A knowledge graph has three primitives:

Primitive	Description	Examples in this KG
Nodes	Objects or concepts	Gene, Protein, Metabolite, Sample, Cohort, Network, Factor, Pathway, Study
Edges	Typed directed relationships	`upregulated`, `gene_part_of`, `gene_contributes_to_factor`, `kegg_pathway_enriched_in`
Properties	Metadata on nodes and edges	`adj_pvalue`, `abs_log2fc`, `network_membership` (kME), `mofa_weight`, `correlation_factor`

3.3 Data Model

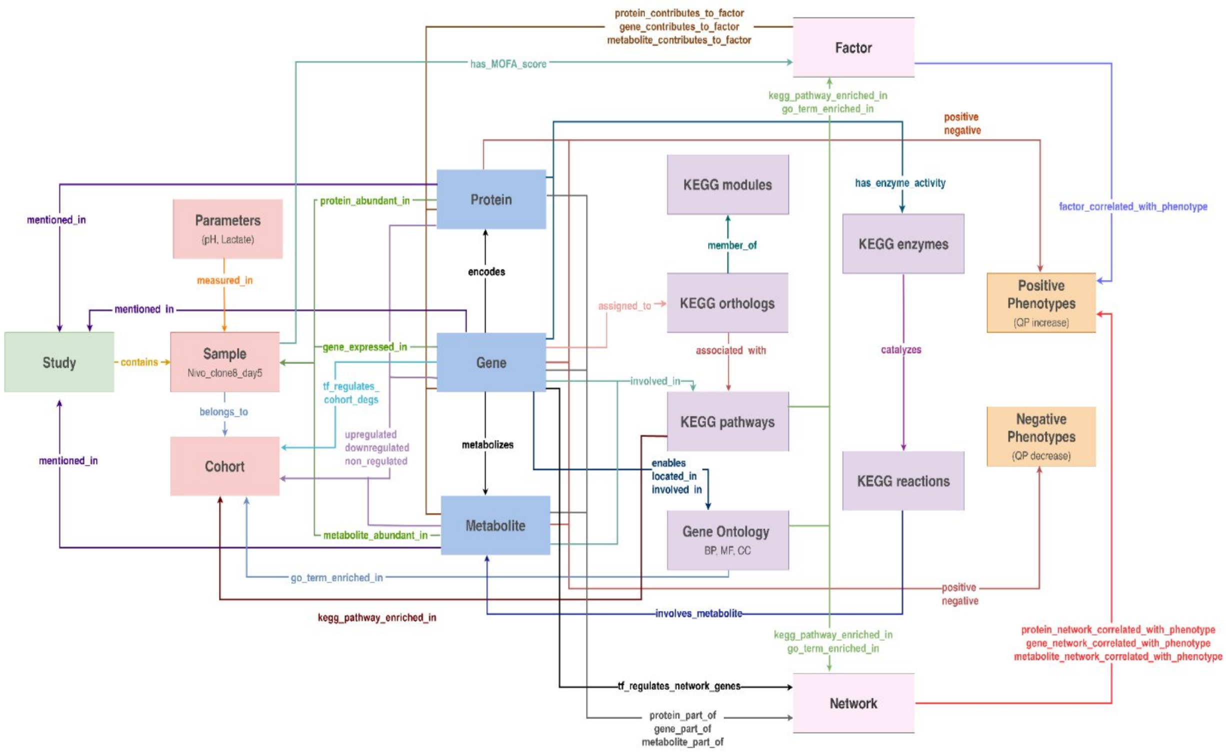

Figure 3.1: **Figure 1.1** The BMS KG data model. Nodes (coloured boxes) represent entity types; labelled arrows are named edge types. Blue nodes are molecular features (Gene, Protein, Metabolite); pink nodes are experimental design entities (Sample, Cohort, Study, Parameters); purple/tan nodes are analytical results (Network, Factor, Phenotypes); grey-purple nodes are pathway and functional annotations.

Every node type in the KG and its role:

Node	Role
Gene	CHO (*Cricetulus griseus*) gene (NCBI Gnomon). Primary join key across all omics layers. Carries human/mouse/rat ortholog IDs.
Protein	Protein detected in BMS proteomics (RefSeq accession). Linked to its encoding Gene via `encodes`.
Metabolite	Small molecule from BMS metabolomics (ChEBI ID). Carries KEGG, HMDB, PubChem, BMS-internal fields.
Sample	One clone × one harvest day × one study. Leaf-level experimental unit.

Node	Role
Cohort	Pairwise comparison between two sample groups (e.g., high vs. low producers). Anchor for all DEG/DAA and enrichment results.
Study	A CHO productivity experiment — internal BMS or published public dataset.
Network	A WGCNA co-expression or co-abundance module (gene, protein, or metabolite layer).
Factor	A MOFA latent factor capturing coordinated variation across omics layers.
Pathway	KEGG pathway (CHO-specific, organism code `cge`).
GO Term	Gene Ontology term — Biological Process (BP), Molecular Function (MF), or Cellular Component (CC).
Positive / Negative Phenotype	Desirable or undesirable bioprocess outcome (e.g., `increased_specific_productivity_qp`).
Parameter	Quantitative bioprocess measurement (e.g., qP, IVCD, lactate) stored per sample.
KEGG Enzyme / Reaction / Ortholog / Module	KEGG metabolic network entities enabling mechanistic pathway traversal.

3.3.1 Key Edge Types

Edge	Connects	Key properties
`upregulated` / `downregulated` / `non_regulated`	Gene/Protein/Metabolite → Cohort	`abs_log2fc`, `adj_pvalue`, `direction`, `sampling_type`
`gene_expressed_in`	Gene → Sample	`abs_log2_tpm`, `direction`
`protein_abundant_in`	Protein → Sample	`abs_log2_abundance`, `direction`
`metabolite_abundant_in`	Metabolite → Sample	`abs_log2_abundance`, `sampling_type` (pellet / supernatant)
`gene_part_of`	Gene → Network	`network_membership` (kME), `gs_qp`, `gs_ivcd`, `gs_titre`, `is_hub`
`gene_contributes_to_factor`	Gene → Factor	`mofa_weight`
`positive` / `negative`	Gene/Protein/Metabolite → Phenotype	`correlation_factor`, `pvalue`, `cause`, `exact_sentence`, `causal_weight`
`kegg_pathway_enriched_in`	Pathway → Cohort/Network/Factor	`enrichment_score`, `padj`, `analysis_type`, `lead_genes`

Edge	Connects	Key properties
`go_term_enriched_in`	GO Term → Cohort/Network/Factor	`enrichment_score`, `padj`, `source_database`
`tf_regulates_cohort_degs`	Gene (TF) → Cohort	`padj`, `direction` (UP/DOWN), `tf_symbol`, `target_ncbi_ids`
`factor_correlated_with_phenotype`	Factor → Phenotype	`correlation`, `direction`
`gene_network_correlated_with_phenotype`	Network → Phenotype	`pearson_correlation`, `fdr`, `direction`
`measured_in`	Parameter → Sample	`bms_bioprocess_parameter_value`, `bms_bioprocess_parameter_unit`

3.4 What Data Lives in the KG?

3.4.1 BMS In-House Data

Study 1 — full multi-omics at Days 3, 5, 9, 11, 14:

Molecule	Clones	High qP	Low qP	Omics layers
C144	5	2	3	Transcriptomics · Proteomics · Metabolomics
Nivo	3	1	2	Transcriptomics · Proteomics · Metabolomics

Study 2 — proteomics only at Days 5, 7, 9, 11:

Molecule	Clones	High qP	Low qP
Milvaxin	6	3	3
COVID2	6	3	3
Nivo	3	1	2
FXIaR	6	3	3

Bioprocess parameters measured in Study 1: qP · titre · IVCD · lactate · O · ammonia · glucose · cell density.

3.4.2 Public Data

- 11 transcriptomics datasets from 10 publications
- 3 proteomics public datasets
- ~290 publications on genetic perturbation → productivity outcomes

3.4.3 External Databases

NCBI Gene · UniProt · ChEBI · KEGG · Gene Ontology · PubMed · Cellosaurus · NCIT

3.5 Cohort Naming Conventions

The KG holds **883 cohorts** across five categories. Always name your cohort explicitly when querying for DEG or enrichment results.

Category	Count	Example
Productivity-Based (all clones & days)	6	`Nivo_study1_all_high_vs_Nivo_study1_al`
Clone-Wise (all days)	36	`Nivo_clone8_study1_all_vs_Nivo_clone49_`
Temporal (same clone, different days)	190	`Nivo_clone49_day11_study1_vs_Nivo_clone`
Temporal Cross-Clone	630	`Nivo_clone8_day11_study1_vs_Nivo_clone4`
Public (high vs. low / non-producer)	19	`GSE295486_high_producer_vs_GSE295486_no`

> **Tip**
>
> Use **Productivity-Based cohorts** for global "which genes drive high productivity?" questions. Use **Clone-Wise** or **Temporal** cohorts for clone-specific or time-resolved analysis.

3.6 How to Phrase Natural-Language Queries

Which genes are upregulated in cohort Nivo_study1_all_high_vs_Nivo_study1_all_low with adj_pvalue < 0.05 and abs_log2fc > 2?

Which proteins are positively correlated with qP (Spearman r > 0.5) in Study 1, across all molecules?

> **Tip**
>
> If a query returns nothing, relax one constraint at a time — e.g., lower the cohort consistency requirement from 7/10 to 5/10, or widen the fold-change threshold.

4 Internal Data Exploration

Browsing BMS data, differential expression, co-expression networks, multi-omics integration, pathway enrichment, and transcription factor analysis

4.1 Overview

This chapter covers everything you can learn from BMS in-house data. It is organised into six layers, each building on the previous:

4.2 A · Data Browsing

What samples are available for the Nivo molecule in Study 1?

Which days have both transcriptomics and proteomics measurements in Study 1?

Show qP, titre, and IVCD values across all Nivo clones in Study 1.

Use the molecule filter to restrict any query to a single drug candidate (`Nivo` or `C144`).

4.3 B · Differential Expression & Abundance

4.3.1 What is DEG/DAA?

Differential Expression Analysis (DEG) identifies genes whose mRNA levels differ significantly between two sample groups — in this KG, typically high vs. low producers. **Differential Abundance Analysis (DAA)** applies the same logic to proteins and metabolites.

Results are stored as directed edges from each molecular feature to a Cohort node, with properties capturing the magnitude and significance of the difference.

Omics layer	Method	Covariate	Normalisation
Transcriptomics (DEG)	DESeq2	Time	TPM
Proteomics (DAA)	T-test	Time	Median central log
Metabolomics (DAA)	T-test	Time	Median central log

4.3.2 Querying DEG/DAA Results

Which genes are upregulated in cohort Nivo_study1_all_high_vs_Nivo_study1_all_low with adj_pvalue < 0.05 and abs_log2fc > 2?

Show genes downregulated in high vs. low C144 producers in Study 1, with abs_log2fc > 1.5.

Which proteins are differentially abundant in high vs. low Nivo producers in Study 1?

Show differentially abundant metabolites in the pellet fraction for high vs. low C144 producers.

Which genes are consistently upregulated in at least 7 out of 10 high vs. low producer cohorts in Study 1?

The multi-filter shortlisting funnel (Study 1):

Filter step	Genes remaining
All DEGs across 232 cohorts	11,481
adj_pvalue < 0.05 and abs_log2fc > 2	5,682
Directional consistency — RNA and protein both up or both down	775
Upregulated in high producers	330
Concordant RNA–protein regulation and Spearman r > 0.5 with qP	**63**

4.3.3 How to Interpret DEG/DAA Results

Output column	Source edge property	Meaning
`abs_log2fc`	`upregulated.abs_log2fc`	Absolute log fold-change (cohort A vs. B). Higher = larger difference.
`direction`	`upregulated.direction`	`positive` = higher in cohort A (high producers); `negative` = higher in cohort B
`adj_pvalue`	`upregulated.adj_pvalue`	BH-corrected p-value. Use < 0.05 as significance threshold.
`pvalue`	`upregulated.pvalue`	Nominal p-value. Prefer adj_pvalue for filtering.

Output column	Source edge property	Meaning
`sampling_type`	Metabolomics edges only	`pellet` = intracellular; `supernatant` = extracellular fraction

> **Tip**
>
> For robust targets, filter for genes with consistent direction in 7/10 high-vs-low cohorts AND both RNA and protein showing the same directional change. The 63-gene shortlist above is a reliable starting point.

4.4 C · Co-expression Networks (WGCNA)

4.4.1 What is WGCNA?

Weighted Gene Co-expression Network Analysis (WGCNA) groups genes, proteins, or metabolites that vary together across samples into **modules**. Each module is summarised by its **Module Eigengene (ME)** — the first principal component of all members — which is then correlated with bioprocess traits (qP, titre, IVCD) to identify productivity-associated modules.

The pipeline runs in five steps:

> **Published precedent:** Melville et al. (2011, *Journal of Biotechnology*) applied WGCNA to 295 microarrays from 121 CHO cultures and identified six transcriptional modules correlated with qP, growth rate, and viability — directly validating this approach for CHO productivity analysis.

In the KG, modules are stored as **Network nodes** (e.g., `network_2_gene`, `network_1_protein`). Module membership is stored on `gene_part_of`, `protein_part_of`, and `metabolite_part_of` edges.

4.4.2 Querying WGCNA Results

Which gene WGCNA networks (modules) are positively correlated with qP in Study 1?

List all genes in network_2_gene (WGCNA module M1) with their network_membership (kME) scores, sorted descending.

What are the top 10 hub genes (is_hub = true) in network_2_gene?

Show genes in network_2_gene with gs_qp > 0.5, ranked by descending network_membership.

Which protein WGCNA modules correlate with IVCD in Study 1?

Show metabolite module members with significant correlation to qP, ranked by gs_qp.

4.4.3 How to Interpret WGCNA Results

Output column	Source edge property	Meaning
network_membership	gene_part_of.network_membership	kME — how strongly a gene tracks its module eigengene (0–1; closer to 1 = stronger hub)
gs_qp	gene_part_of.gs_qp	Gene significance for qP — Pearson r of gene expression with qP across samples
gs_ivcd	gene_part_of.gs_ivcd	Gene significance for IVCD
gs_titre	gene_part_of.gs_titre	Gene significance for product titre
gs_qp_pval	gene_part_of.gs_qp_pval	P-value for gs_qp — use < 0.05 for significance
sig_qp / sig_ivcd / sig_titre	Boolean flags	true = gene–trait association is statistically significant
is_hub	gene_part_of.is_hub	true = high intramodular connectivity; hub gene
pearson_correlation	gene_network_correlated_with_phenotype	Module eigengene vs. phenotype Pearson r
fdr	Same edge	BH-corrected FDR for module–phenotype correlation

> **Note**
>
> **Key modules identified in Study 1:**
>
> - `network_2_gene` (M1 Pro-productivity) — ER processing / UPR enriched. ME positively correlated with qP.
> - `network_18_gene` (M18 Anti-productivity) — Cell cycle / DNA replication enriched. ME negatively correlated with qP.

4.5 D · Multi-Omics Factor Analysis (MOFA)

4.5.1 What is MOFA?

Multi-Omics Factor Analysis (MOFA) is an unsupervised dimensionality-reduction method that integrates transcriptomics, proteomics, and metabolomics simultaneously to identify **latent factors** — hidden biological programs that explain variation across samples.

Each factor is a weighted linear combination of features across all three omics layers. Features with large weights drive that factor; the factor score for each sample reflects how strongly the sample expresses that biological program.

Significant factors and their biological meaning:

Factor	Correlation pattern	Interpretation
Factor 1	qP +0.848 · IVCD +0.860 · Titre +0.904	Full-spectrum productivity signal — the most actionable factor
Factor 2	qP positive · IVCD negative	High efficiency per cell; not driven by biomass accumulation
Factor 3	Proteomics-only	Post-translational regulatory axis decoupled from transcription
Factor 4	Transcriptomics-only	Transcriptional program not yet reflected in protein or metabolite levels
Factor 5	Titre and qP strongly negative	Pure metabolic suppressor acting on secretory machinery

4.5.2 Querying MOFA Results

Which genes have the highest positive MOFA Factor 1 weight in Study 1?

Show proteins with MOFA Factor 1 weight above 0.3.

Which metabolites contribute most to Factor 5? What pathways are enriched in those metabolites?

Show per-sample MOFA Factor 1 scores for all Study 1 Nivo samples and overlay with qP values.

Show genes that are consistently upregulated in high producers (7/10 cohorts), members of a pro-productivity WGCNA network, AND have positive MOFA Factor 1 weight. This is the triple-overlap pro-productivity shortlist.

4.5.3 How to Interpret MOFA Results

Output column	Source edge property	Meaning
`mofa_weight`	`gene_contributes_to_factor.mofa_weight`	MOFA loading weight. Positive = co-varies positively with factor; negative = inversely. Typical range: −1 to +1.
`entity_type`	Same edge	`gene`, `protein`, or `metabolite` — which omics layer this feature belongs to

Output column	Source edge property	Meaning
`correlation`	`factor_correlated_with_phenotype.correlation`	Spearman correlation between factor scores and phenotype values across samples
`direction`	Same edge	`positive` / `negative` — whether high factor scores associate with high or low phenotype
`mofa_score`	`has_mofa_score.mofa_score`	Per-sample factor loading score (typically −5 to +5)

> **!** Important
>
> **Factor 1 is the most actionable.** Genes with high positive Factor 1 weights are pro-productivity candidates for overexpression. Genes with high negative weights are anti-productivity candidates for knockout.

4.6 E · Pathway & Functional Enrichment

4.6.1 What is Pathway Enrichment?

Pathway enrichment analysis tests whether a set of genes or metabolites contains more members of a given biological pathway than expected by chance. Results are **pre-computed and stored in the KG** as edges from pathway nodes to cohort, network, or factor nodes — you query the results, not run the analysis.

Method	Input	What it detects
GSEA (Gene Set Enrichment Analysis)	Ranked gene list (by log2FC or MOFA weight)	Coordinated shifts across the full ranked list
ORA (Over-Representation Analysis)	Binary DEG or module member list	Pathways with more hits than expected by chance

Both are applied to DEG cohorts, WGCNA modules, and MOFA factors. Databases: **KEGG Pathways** and **Gene Ontology** (BP, MF, CC).

4.6.2 Querying Pathway Enrichment

On DEG cohorts:

Which KEGG pathways are enriched (GSEA) in cohort Nivo_study1_all_high_vs_Nivo_study1_all_low?

Show GO Biological Process terms enriched in downregulated genes of C144 high vs. low producers.

On WGCNA modules:

What KEGG pathways are enriched in gene network network_2_gene?

Show GO terms enriched in the anti-productivity gene network network_18_gene.

On MOFA factors:

Which pathways are enriched in the top-weighted genes for MOFA Factor 1?

Show KEGG pathway enrichment for MOFA Factor 5, separating positive and negative weight metabolites.

Cross-molecule comparison:

Compare enriched KEGG pathways between Nivo and C144 high vs. low producer cohorts. Which are shared and which are molecule-specific?

4.6.3 How to Interpret Enrichment Results

Output column	Source edge property	Meaning
`enrichment_score`	`kegg_pathway_enriched_in.enrichment_score`	NES for GSEA, fold-enrichment for ORA. Positive = enriched in upregulated / positive-weight genes.
`padj`	Same edge	BH-corrected p-value. Use < 0.05.
`overlap`	Same edge	Fraction of pathway genes in the input set. Higher = more pathway coverage.
`analysis_type`	Same edge	`GSEA`, `ORA_upregulated`, `ORA_downregulated`, `ORA_Negative`
`lead_genes`	Same edge	Gene symbols driving the enrichment signal (GSEA leading edge or ORA hits)
`sampling_type`	Metabolomics edges only	`pellet` (intracellular) or `supernatant` (extracellular)
`source_database`	`go_term_enriched_in`	`GO_BP`, `GO_MF`, or `GO_CC`

4.6.4 KEGG Metabolic Network Traversal

The KG also stores the full KEGG metabolic network, enabling mechanistic gene-to-reaction queries:

Which KEGG pathway does gene LDHA belong to?

Find all genes in the glycolysis KEGG pathway that are differentially expressed in high vs. low Nivo producers.

Trace LDHA to its enzymatic activity and the metabolic reactions it catalyses.

4.7 F · Transcription Factor Analysis

4.7.1 What is TF Enrichment?

Transcription factor (TF) enrichment identifies upstream regulators whose target genes are over-represented in a DEG set or WGCNA module. It answers: *which TFs are driving the gene expression changes we observe in high producers?*

The analysis uses **gseapy (enrich)** against the **ChEA 2022** database of curated TF–target relationships (from ChIP-seq experiments). Results are stored as edges from TF Gene nodes to Cohort or Network nodes.

Edge type	Meaning
`tf_regulates_cohort_degs`	TF whose targets are enriched in a cohort's DEG set
`tf_regulates_network_genes`	TF whose targets are enriched in a WGCNA module's gene set

Available for: **Study 1 cohorts** and **WGCNA gene modules**.

4.7.2 Querying TF Results

Which transcription factors regulate genes upregulated in cohort Nivo_study1_all_high_vs_Nivo_study1_all_low?

Show TFs with padj < 0.01 whose targets are enriched among upregulated DEGs in C144 high producers.

Which TFs regulate genes in pro-productivity WGCNA network network_2_gene?

What TFs control the anti-productivity gene network network_18_gene?

Trace gene HSPA5 upstream: what transcription factors regulate it in high-producer cohorts?

Which TFs have the broadest target gene scope across all pro-productivity cohorts in Study 1? Identify master regulators.

4.7.3 How to Interpret TF Results

Output column	Source edge property	Meaning
`tf_symbol`	`tf_regulates_cohort_degs.tf_symbol`	Standardised gene symbol of the TF (e.g., `E2F4`, `NFE2L2`)
`direction`	Same edge	`UP` = TF targets enriched in upregulated DEGs; `DOWN` = in downregulated DEGs
`padj`	Same edge	BH-corrected p-value for TF enrichment significance
`overlap`	Same edge	Fraction of TF targets in the DEG set
`target_genes`	Same edge	Semicolon-delimited list of TF target gene symbols in the overlap
`tf_term_original`	Same edge	Full ChEA term (e.g., `E2F4 20460467 ChIP-Seq MCF7 Human`) — shows source organism and assay
`tf_database`	Same edge	Always `ChEA_2022` in this KG

5 Public Literature Evidence

What published studies say about genes, proteins, metabolites, and CHO productivity

5.1 Overview

The KG integrates cause-effect relationships curated from ~**290 publications** on genetic perturbation in the context of CHO cell productivity. This chapter explains how to query that evidence layer.

5.2 How Literature Evidence is Structured

Publications were scouted from ~45,000 CHO-related papers through a two-stage filter:

Filter 1 — category classification:

- **(A)** Gene/protein/metabolite target manipulation and cell engineering studies — *included*
- **(B)** Bioprocess optimisation
- **(C)** Other studies

Filter 2 — inclusion/exclusion:

- Include: productivity insights through target manipulation or omics-driven cell-line engineering
- Exclude: not directly relevant to target manipulation impacting productivity

Result: ~**290 publications** ingested, describing overexpression (OE), knockdown (KD), and knockout (KO) interventions and their effects on productivity phenotypes.

Each extracted relationship is stored as a **directed edge** (`positive` or `negative`) from a Gene/Protein/Metabolite node to a Positive or Negative Phenotype node, carrying the following properties:

Property	Description
`cause`	Free-text description of the causal experimental manipulation (e.g., "BiP co-expression")
`effect_entity`	Free-text description of the affected biological measurement (e.g., "volumetric titer")

Property	Description
`causal_weight`	Numeric confidence weight of the causal claim (0 to 1; higher = stronger evidence)
`exact_sentence`	Verbatim supporting sentence(s) from the source publication
`quantitative_value`	Quantitative result reported in the literature (e.g., "20% increase")
`gene_source_organism`	Scientific name of the source organism (e.g., `Cricetulus griseus`, `Homo sapiens`)
`study_id`	PMID or study identifier of the source publication
`source`	`public` for all literature-derived edges
`pvalue`	P-value for the reported association (where available)

> **Note**
>
> A gene can be **positively associated** with a positive phenotype (higher expression leads to better outcome) or **negatively associated** (lower expression or knockout leads to better outcome). Both types are equally important and are included in results unless you explicitly filter by direction.

5.3 Key Effect Entities

The KG tracks effects across four biological dimensions:

Productivity: specific_productivity_qp · product_titer · volumetric_productivity · secretion_rate · product_yield · production_stability

Growth: viable_cell_density_vcd · cell_viability · integral_viable_cell_density_ivcd · specific_growth_rate · apoptosis_rate · cell_cycle_phase_distribution · cloning_efficiency

Metabolic: lactate_production · lactate_consumption · ammonia_production · glucose_consumption · glutamine_metabolism · redox_balance · metabolic_flux · lipid_metabolism_rate

Quality: glycosylation_profile · fucosylation_level · sialylation_level · product_purity · protein_fragmentation · aggregation_level · host_cell_protein_content · product_bioactivity

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5.4 Querying Literature Evidence

5.4.1 Finding Genes Linked to Productivity

Which genes are linked to increased specific productivity (qP) in published CHO studies?

Show all genes with published evidence for improving product titer in CHO cells, including the cause and exact_sentence.

5.4.2 Looking Up a Specific Gene

What does the literature say about HSPA5 and CHO productivity? Include manipulation type, effect, causal_weight, and PMID.

5.4.3 Filtering by Phenotype and Direction

Which genes, when overexpressed, show a positive association with product titer in CHO?

Show genes whose knockout reduces protein fragmentation. Include causal_weight and exact_sentence.

5.4.4 Filtering by Source Organism

Show only CHO-specific (Cricetulus griseus) literature evidence for HSPA5.

Are there human or mouse studies for FOXO1 that might be relevant to CHO productivity? Show gene_source_organism for each result.

5.4.5 Retrieving Supporting Sentences

Show the exact_sentence values from publications linking HSPA5 overexpression to increased titre.

5.4.6 Genes with Experimental Manipulation

Which genes have been experimentally overexpressed in CHO cells with a positive effect on qP? Show PMID and causal_weight, sorted by causal_weight descending.

5.5 Worked Example: HSPA5 (GRP78/BiP)

Literature

HSPA5 encodes the ER chaperone GRP78/BiP, a master regulator of the unfolded protein response. The KG contains 7 published associations linking HSPA5 to positive productivity phenotypes across 3 independent CHO studies:

Phenotype	Cause	Effect entity	Study (PMID)
Increased specific productivity (qP)	BiP co-expression	Specific production rate (qP)	24081924
Increased product titer	BiP co-expression	Volumetric titer	24081924
Increased product titer	Co-expression of BiP and CypB	Sp35Fc titer (+20%)	26126657
Increased volumetric productivity	GRP78 overexpression	M-CSF productivity	1373378
Decreased apoptosis rate	GRP78 overexpression during H O exposure	Cleaved caspase-3 levels	32624757
Increased product titer	GRP78 overexpression	TfR-Ab concentration	32624757
Increased cell viability	GRP78 overexpression (serum-free SFM4CHO)	Cell viability (33.87% vs 3.23% at Day 11)	32624757

Selected supporting evidence sentences:

> PMID 24081924: "BiP significantly improved volumetric titer of DTE MAb E, mediated specifically via an increase in qP."
>
> PMID 26126657: "A combination of two genes, BiP and CypB, increased Sp35Fc titer by 20% compared to that obtained with either BiP or CypB alone."
>
> PMID 32624757: "ELISA assay showed that the concentrations of TfR-Ab in the supernatant of CHO-GRP78 cells were higher than those in CHO cells at any time points." Cell viability at Day 11: CHO-GRP78 cells 33.87% vs CHO cells 3.23%.

Mechanism: UPR activation expands ER folding capacity, improves secretory efficiency, and reduces apoptosis under stress conditions — all contributing to higher titre and qP.

5.6 Finding Genes Mentioned in a Specific Publication

Which genes are mentioned in study with PMID 32624757?

Show all cause-effect relationships extracted from PMID 38898154.

6 Cross-Evidence Integration

Concordance and discordance between BMS internal data and published literature

6.1 Overview

The highest-confidence engineering candidates are those supported by **both** BMS internal omics data and independent published literature. This chapter explains how to find, interpret, and resolve agreement — or disagreement — between these two evidence streams.

6.2 Why Cross-Evidence Analysis Matters

Internal data alone surfaces many differentially expressed genes — not all are validated targets. Literature alone reflects findings in other cell lines, products, and contexts. When both sources agree in direction, confidence rises substantially.

The KG stores BMS omics results and literature associations as separate edge types on the same Gene nodes. A single query can traverse both simultaneously.

6.3 Checking if a Literature Gene Appears in BMS Data

Is HSPA5 differentially expressed in any BMS Study 1 high vs. low producer cohorts? Show adj_pvalue and abs_log2fc.

Is HSPA5 a member of a WGCNA gene network correlated with productivity in Study 1? Show network_membership and gs_qp.

Does HSPA5 have a significant MOFA Factor 1 weight in Study 1?

What is the Spearman correlation of HSPA5 expression with qP in Study 1?

6.4 Direction Concordance

The most important check — does the **direction** of the BMS signal agree with the published manipulation effect?

BMS signal	Published effect	Verdict
Gene upregulated in high producers	OE → ↑ qP	Concordant — strong OE candidate
Gene downregulated in high producers	KO → ↑ titer	Concordant — strong KO candidate
Gene upregulated in high producers	OE → ↓ qP	Discordant — investigate context
Gene not DE in BMS data	OE → ↑ qP	BMS data silent — check WGCNA and MOFA layers

For genes with published evidence of OE showing positive qP effect, which are also upregulated in high vs. low producers in BMS Study 1?

Show genes where BMS data shows upregulation in high producers AND literature shows OE → increased titer. Include abs_log2fc, correlation_factor, and exact_sentence.

6.5 Explaining Discordance

When BMS and literature disagree, common explanations:

- **Source organism** — study was in human or mouse cells; CHO physiology differs. Check `gene_source_organism` on the edge.
- **Product type** — effect seen for mAb production may not hold for fusion proteins or DTE proteins
- **Cell line background** — CHO-K1, CHO-DG44, and CHO-S have regulatory differences
- **Culture conditions** — fed-batch vs. batch; different temperature shifts or pH profiles
- **Timepoint** — perturbation effect measured at a different culture stage

Gene X is upregulated in BMS high producers but literature shows OE → decreased qP. Are those literature studies from CHO or human/mouse cells? Show gene_source_organism for each association.

6.6 Genes with Multi-Omics + Literature Support

Show genes that are: (1) consistently upregulated in high producers across 7/10 BMS Study 1 cohorts, (2) members of a pro-productivity WGCNA network, (3) have positive MOFA Factor 1 weight, AND (4) have published evidence linking them to increased qP or titer.

6.7 Mapping BMS Data to Public Omics

Is HSPA5 differentially expressed in any of the public high-vs-low producer omics datasets in the KG?

Which genes from the 63-gene BMS shortlist are also differentially expressed in at least 3 public CHO datasets?

6.8 Genes Unique to BMS Data (No Literature Precedent)

Show genes consistently upregulated in high producers in BMS Study 1 that are NOT mentioned in any ingested publication. These are potential novel targets.

7 Hypothesis Testing

Interrogating the KG when you arrive with a prior gene, pathway, or mechanism

7.1 Overview

This chapter is for users who arrive at the KG with a **prior** — a gene of interest from a paper, a pathway hypothesis, or a specific biological mechanism to test. The KG lets you rapidly check how much evidence exists across all analytical layers.

7.2 I Have a Gene — Multi-Layer Audit

Use this as your default starting point whenever you have a gene of interest.

Tell me everything the KG knows about gene HSPA5: differential expression in high vs. low producers, WGCNA network membership, MOFA weight, Spearman correlation with qP and titre, and published evidence.

The KG will return evidence from:

1. DEG/DAA Is it DE? In how many cohorts? In which direction?
2. WGCNA Which network does it belong to? What is kME? Is the network correlated with productivity?
3. MOFA Does it have a significant Factor 1 weight?
4. Spearman correlation with qP, titre, and IVCD directly.
5. Literature Has it been manipulated in CHO cells? With what effect?

7.3 I Have a Pathway — Which Genes in it Are Productivity-Associated?

Which genes in the "Protein processing in endoplasmic reticulum" KEGG pathway are differentially expressed in BMS Study 1 high vs. low producer cohorts?

Show all genes in the glycolysis KEGG pathway with positive Spearman correlation to qP in Study 1.

7.4 I Have a Gene List — Check Against All Evidence Layers

For each of these genes: HSPA5, DNAJB9, MERTK, FICD, ZCCHC24 — show DEG status in Nivo high vs. low producers, WGCNA network, MOFA Factor 1 weight, and literature evidence.

7.5 I Have a Phenotype — What Genes Are Most Correlated?

Which genes have the strongest positive Spearman correlation with qP in Study 1? Rank descending.

Rank proteins by their Spearman correlation with titre in Study 1, descending.

Which metabolites are most negatively correlated with qP in Study 1?

7.6 Testing Specific Biological Hypotheses

7.6.1 ER Stress / Unfolded Protein Response (UPR)

Prior: High producers activate UPR to expand ER folding capacity.

Show all genes involved in the UPR or ER stress response that are upregulated in high producers in Study 1.

Is "protein processing in endoplasmic reticulum" enriched in the pro-productivity WGCNA network network_2_gene?

Which UPR-related transcription factors (ATF4, ATF6, XBP1) are differentially expressed in high producers in Study 1?

7.6.2 Secretory Pathway

Prior: High producers have enhanced protein secretion machinery.

Are genes in the "protein export" and "antigen processing" pathways upregulated in high producers in Study 1?

Show the top co-expressed genes with HSPA5 in WGCNA network network_2_gene.

7.6.3 Metabolic Bottleneck (Lactate / Ammonia / ROS)

Prior: Runaway lactate accumulation limits productivity in low producers.

Which metabolites differ significantly between high and low producers in Study 1? Highlight lactate.

Show genes in the lactate metabolism pathway with negative Spearman correlation to titre in Study 1.

Are there published interventions (gene KO or OE) that reduce lactate accumulation in CHO cells? Show cause, effect_entity, and exact_sentence.

Which published studies consistently link apoptosis-resistance interventions (combined knockouts) to lower lactate production? Show exact_sentence and study_id.

7.6.4 Chaperone / Difficult-to-Express (DTE) Proteins

Prior: DTE proteins benefit from co-expression of specific chaperones.

Show all chaperone genes (HSP family, DNAJ family) upregulated in high producers in Study 1.

What publications in the KG link chaperone overexpression to improved productivity of difficult-to-express proteins? Show cause, effect_entity, and exact_sentence.

> **Note**
>
> The KG returns results for DTE-related queries — it uses publication titles and abstracts to identify studies targeting difficult-to-express proteins. Confirmed working as of KG Version 3.

7.6.5 Apoptosis Resistance

Prior: Blocking apoptosis extends culture longevity and increases total titre.

Are anti-apoptotic genes (BCL2 family, BCL-XL) upregulated in high producers in any BMS Study 1 cohort?

Which published studies link apoptosis-resistance interventions (combined knockouts) to reduced lactate and improved productivity? Show exact_sentence.

8 KG Utilities & Navigation

Searching, cross-referencing identifiers, filtering, exporting results, and generating plots

8.1 Overview

This chapter covers the practical mechanics of navigating the KG: searching by identifier, cross-referencing databases, filtering results, exporting data, and generating visualisations.

8.2 Searching by Identifier

Entity	Supported identifiers
Gene	Gene symbol (e.g., `HSPA5`), NCBI Entrez Gene ID (e.g., `100752894`)
Protein	UniProt accession (e.g., `P11021`), RefSeq accession (e.g., `XP_027251020.1`)
Metabolite	ChEBI ID (e.g., `CHEBI:16651`), metabolite name, KEGG Compound C-number, HMDB ID
Pathway	KEGG pathway ID (e.g., `cge04141`), pathway name
GO Term	GO term ID (e.g., `GO:0006457`), GO term name

Find gene HSPA5. Show its NCBI Gene ID, UniProt accession of its encoded protein, and all associated KEGG pathways.

Search for metabolite lactate using its ChEBI ID CHEBI:16651. Show all WGCNA networks it belongs to and its correlation with qP.

8.3 Cross-Referencing Identifiers

What is the UniProt accession for the protein encoded by HSPA5?

What CHO gene symbol corresponds to NCBI Gene ID 100752894?

What is the KEGG ortholog (K-number) for gene LDHA in CHO cells?

8.4 Finding Orthologs

CHO genes are mapped to human, mouse, and rat orthologs via the `human_id`, `mouse_id`, and `rat_id` fields on Gene nodes. KEGG Ortholog (`assigned_to`) edges provide functional ortholog groupings.

Find the human ortholog of CHO gene LOC100760815. What is its NCBI Gene ID?

Which human genes are orthologs of the top 10 hub genes in WGCNA network_2_gene?

8.5 Filtering Results

8.5.1 By Study Source

Show DEG results for HSPA5 from BMS internal studies only (source = 'bms').

Show only public omics datasets with DEG results for KCTD14.

8.5.2 By Molecule

Filter WGCNA network enrichment results to Nivo samples only (sample_source = 'Nivo_study1').

Show MOFA Factor 1 weights for genes, using only C144 Study 1 samples (sample_source = 'C144_study1').

Threshold	value	Edge property

8.5.3 Statistical Thresholds

Threshold	value	Edge property
DEG adjusted p-value	< 0.05	`adj_pvalue`
DEG absolute log FC	> 1	`abs_log2fc`
Spearman correlation	$\|r\| > 0.5$	`correlation_factor`
MOFA weight (top features)	$>$ 80th percentile	`mofa_weight`
WGCNA module membership	kME > 0.5	`network_membership`
TF enrichment	padj < 0.05	`padj`
Pathway enrichment	padj < 0.05	`padj`

8.6 Plotting Routines

The KG can generate a variety of visualisations directly from query results. Ask Co-Scientist to produce plots alongside tabular output.

What kinds of plots can you generate for KG query results?

Plots are generated directly by the Co-Scientist from KG data. The following types are available, each backed by specific KG edge properties:

Plot type	KG data source	When to use
KG subgraph / network view	Any edge type (e.g., `upregulated`, `positive`, `gene_part_of`, metabolic chains `gene→kegg_ortholog→kegg_enzyme→kegg_reaction→metabolite`)	Visualise relationships around a specific gene, protein, or metabolite; trace metabolic paths
Volcano / scatter plot	`upregulated` / `downregulated` edges: x = `abs_log2fc`, y = -log10(`pvalue`), colour by edge type	DEG/DAA — identify significant differentially expressed features per cohort
Expression heatmap	`gene_expressed_in.abs_log2_tpm`, `protein_abundant_in.abs_log2_abundance`, `metabolite_abundant_in.abs_log2_abundance`; faceted by `harvest_day`, `clone_id`, `molecule`	Compare expression profiles across samples, clones, or molecules

Plot type	KG data source	When to use
Enrichment bar / lollipop	`go_term_enriched_in` and `kegg_pathway_enriched_in`: y = -log10(`padj`)	Top enriched GO terms or KEGG pathways per cohort, network, or factor
Enrichment dot / bubble plot	Same enrichment edges: y = term, x = `enrichment_score`, size = `overlap`, colour = -log10(`padj`)	Compare enrichment across multiple terms simultaneously
WGCNA module–phenotype bars	`gene/protein/metabolite_network_correlated_with_phenotype`: `pearson_correlation` with `fdr`, coloured by `direction`	Which modules correlate with pro- or anti-productivity
MOFA factor correlation bars	`factor_correlated_with_phenotype.correlation`, coloured by `direction`	Which MOFA factors correlate with qP, titre, IVCD
MOFA loadings plot	`mofa_weight` from `gene/protein/metabolite_contributes_to_factor`	Top features driving each factor, per omics layer
Pathway treemap / sunburst	`pathway.top_level_mapping_broad` and `top_level_mapping_subcategory`	Hierarchical breakdown of enriched pathways by KEGG BRITE category
Bioprocess time-series	`measured_in.bms_bioprocess_parameter_value` over `Sample.harvest_day`, faceted by `molecule` or `qp_group`	qP, titre, lactate, glucose trends across culture days
Cohort summary stacked bar	Count of `upregulated/downregulated` entities per cohort; or top enriched pathway by `padj`	Overview of DE landscape across multiple cohorts

Show a scatter plot of HSPA5 expression vs. qP across all Study 1 samples.

Generate a bar chart of GSEA enrichment scores for the top 10 enriched KEGG pathways in cohort Nivo_study1_all_high_vs_Nivo_study1_all_low.

Show a heatmap of the top 20 upregulated genes in high vs. low Nivo producers across all Study 1 samples.

> **Note**
>
> Plot availability depends on the Python backend connected to the KG session. Network plots, heatmaps, and scatter plots are consistently available. For specific plot types, ask Co-Scientist first — it will confirm what is currently supported.

8.7 Exporting Query Results

Results can be exported from the session toolbar after any query returns a table or plot.

8.7.1 Exporting Tables

Export all WGCNA network_2_gene members with network_membership, gs_qp, gs_titre, and is_hub as a CSV.

Export the top 50 genes by positive MOFA Factor 1 weight across transcriptomics, proteomics, and metabolomics layers.

8.7.2 Exporting Plots

After a plot is generated in the chat:

8.8 Tips for Effective Queries

- **Be specific about entity type** — say “gene”, “protein”, or “metabolite” explicitly
- **Name the cohort** — include molecule, clone, and study when relevant
- **State directionality** — “upregulated”, “positive MOFA weight”, “negatively correlated with qP”
- **Include exact thresholds** — numbers rather than “strong” or “significant”
- **Build progressively** — start broad, then add constraints to narrow results
- **Reference modules and factors by name** — `network_2_gene`, `Factor1`, not “the pro-productivity module”

> **Tip**
>
> If results feel off-topic, start a **New Session** (pencil icon, top right) to clear accumulated context, then rephrase your query with more specificity.

9 Glossary

This glossary is organised into four sections matching the main evidence layers in the KG. Use Ctrl+F / Cmd+F to search for a specific term.

9.1 BMS Internal Data Terms

C144 One of the two BMS antibody molecules for which in-house multi-omics data is available in Study 1 (5 clones; Days 3, 5, 9, 11, 14).

CHO (Chinese Hamster Ovary) The cell line used for BMS biopharmaceutical protein production. All BMS in-house data in this KG comes from CHO cells (*Cricetulus griseus*).

Clone A genetically identical population of cells derived from a single parent cell. Each clone in BMS Studies 1 and 2 produces either a high or low level of the target protein. Clone IDs in the KG follow the format `clone8`, `clone49`, `clone94`, etc.

Cohort A pairwise comparison group used for differential expression or abundance analysis. The KG contains 883 cohorts across five categories (see Section 3.5 for naming conventions).

High Producer / Low Producer Classification of a clone based on its specific productivity (qP). High-producer clones secrete more product per cell per unit time. This classification is stored on the `qp_group` field of Sample nodes.

IVCD (Integral Viable Cell Density) The area under the viable cell density–time curve during a culture run. Measures cumulative biomass produced. Unit: cells $\times$ day / mL. Stored as a bioprocess Parameter node; correlations on `positive`/`negative` edges.

Molecule The therapeutic protein being produced — either `Nivo` (nivolumab, an anti-PD-1 mAb) or `C144` in Study 1; Milvaxin, COVID2, Nivo, and FXIaR in Study 2.

Nivo (Nivolumab) Anti-PD-1 monoclonal antibody. One of the two BMS molecules with full multi-omics data in Study 1.

qP (Specific Productivity) The rate of product secretion per cell per unit time. Units: pg/cell/day. The primary productivity phenotype. Stored as a bioprocess Parameter node; Spearman correlations stored on `positive`/`negative` edges from genes/proteins/metabolites.

Sample A single biological replicate — one clone $\times$ one harvest day $\times$ one study (e.g., `Nivo_clone8_day5_study1`). The leaf-level experimental unit in the KG.

Study 1 BMS in-house multi-omics study with transcriptomics, proteomics, and metabolomics. Nivo (3 clones) and C144 (5 clones) at Days 3, 5, 9, 11, 14.

Study 2 BMS in-house proteomics-only study. Milvaxin, COVID2, Nivo, and FXIaR at Days 5, 7, 9, 11.

Titre Total product concentration in the culture supernatant at harvest (mg/L). Reflects both qP and viable cell density over time. Stored as a bioprocess Parameter node.

9.2 Bioinformatics & Analysis Terms

adj_pvalue (Adjusted P-value) P-value corrected for multiple testing using the Benjamini-Hochberg (BH) method. Values below 0.05 are considered statistically significant. Stored on DEG edges (`upregulated`, `downregulated`), enrichment edges, and TF edges.

DAA (Differential Abundance Analysis) Statistical analysis identifying proteins or metabolites that differ significantly in abundance between two conditions. Method: T-test with time as covariate. Stored as `upregulated`/`downregulated` edges from Protein or Metabolite nodes to Cohort nodes.

DEG (Differentially Expressed Gene) A gene whose mRNA expression differs significantly between two conditions. Method: DESeq2 with time as covariate. Stored as `upregulated`/`downregulated` edges from Gene nodes to Cohort nodes.

DTE (Difficult-to-Express) Proteins with complex folding, disulfide-bond, or glycosylation requirements that make them hard to produce at high levels in CHO cells. The KG returns DTE-related results via publication mining.

Factor (MOFA Factor) A latent variable identified by MOFA that captures coordinated variation across omics layers. Factor 1 is the most strongly correlated with productivity phenotypes. Stored as a Factor node; features linked via `gene_contributes_to_factor`, `protein_contributes_to_factor`, `metabolite_contributes_to_factor` edges.

GS (Gene Significance) In WGCNA, the Pearson correlation between a gene's expression and a productivity trait. Stored as `gs_qp`, `gs_ivcd`, `gs_titre` on `gene_part_of` edges. Corresponding p-values: `gs_qp_pval`, `gs_ivcd_pval`, `gs_titre_pval`.

GSEA (Gene Set Enrichment Analysis) Enrichment method using a ranked gene list to detect coordinated pathway shifts. Reports Normalised Enrichment Score (NES). Positive NES = enriched in upregulated genes. Stored on `kegg_pathway_enriched_in` and `go_term_enriched_in` edges with `analysis_type = 'GSEA'`.

Hub Gene A gene with high intramodular connectivity in a WGCNA module — it is the most representative of the module's biological program. Flagged by `is_hub = true` on `gene_part_of` edges.

kME (Module Eigengene Connectivity) In WGCNA, the Pearson correlation between a gene's expression and its module eigengene. Range 0–1; closer to 1 = stronger hub. Stored as `network_membership` on `gene_part_of` edges.

KO (Knockout) Complete elimination of a gene's function, typically via CRISPR-Cas9. Contrast with KD (knockdown, partial reduction) and OE (overexpression, increased expression).

log2FC (Log Fold-Change) The log ratio of expression/abundance in cohort A vs. cohort B. Stored as `abs_log2fc` (absolute value) on DEG/DAA edges. Positive values indicate higher expression in cohort A (high producers in productivity-based cohorts).

ME (Module Eigengene) The first principal component of all features within a WGCNA module. Represents the overall activity pattern of the module. Used to correlate module activity with productivity traits.

MOFA (Multi-Omics Factor Analysis) Unsupervised dimensionality-reduction framework integrating multiple omics layers into shared latent factors. Run with K=10 factors across transcriptomics, proteomics, and metabolomics.

mofa_weight MOFA loading weight for a feature on a given factor. Direction: positive = co-varies positively with the factor; negative = inversely. Magnitude indicates relative importance. Stored on `gene_contributes_to_factor` and equivalent edges.

NES (Normalised Enrichment Score) The GSEA enrichment score normalised for gene set size. Positive = enriched in upregulated genes; negative = enriched in downregulated genes. Stored as `enrichment_score` on enrichment edges.

NLQ (Natural Language Query) A plain-English question typed into the KG interface (Polly Co-Scientist), automatically converted to a Cypher graph query.

ORA (Over-Representation Analysis) Enrichment method testing whether a binary gene list contains more pathway members than expected by chance. Stored with `analysis_type` = `ORA_upregulated`, `ORA_downregulated`, or `ORA_Negative`.

Spearman Correlation Non-parametric rank correlation used to measure association between a molecular feature's abundance and a productivity phenotype. Stored as `correlation_factor` on `positive`/`negative` edges from internal BMS analysis.

TF (Transcription Factor) A protein that binds DNA to activate or repress gene expression. TF enrichment in the KG uses ChEA 2022 (ChIP-seq-based) data. Results stored on `tf_regulates_cohort_degs` and `tf_regulates_network_genes` edges.

TPM (Transcripts Per Million) RNA-seq normalisation unit accounting for sequencing depth and transcript length. Stored as `abs_log2_tpm` on `gene_expressed_in` edges.

UPR (Unfolded Protein Response) A cellular stress response activated when misfolded proteins accumulate in the ER. Associated with pro-productivity gene expression programs in high CHO producers (enriched in WGCNA network_2_gene).

WGCNA (Weighted Gene Co-expression Network Analysis) Network-based method grouping co-expressed genes, proteins, or metabolites into modules correlated with biological traits. The BMS KG contains 20 gene, 5 protein, and 5 metabolite networks.

9.3 Bioprocess Parameter Terms

Ammonia (mM) Metabolic by-product that accumulates in culture and inhibits cell growth at high concentrations. Stored as a Parameter node; measured per sample on `measured_in` edges.

Glucose (g/L) Primary carbon and energy source for CHO cells. Consumption rate reflects metabolic activity. Stored as a Parameter node.

Lactate (mM) Metabolic by-product of glycolysis. Accumulation is associated with low productivity and metabolic inefficiency. A key bioprocess parameter tracked across all Study 1 samples.

O (Dissolved Oxygen) Dissolved oxygen concentration in the bioreactor. Critical for aerobic metabolism and productivity. Measured per sample in Study 1.

VCD (Viable Cell Density) Number of living cells per mL of culture. Used alongside qP to calculate volumetric productivity. Stored as a Parameter node.

9.4 External Database Terms

ChEBI Chemical Entities of Biological Interest. Database providing IDs and annotations for small molecules. ChEBI IDs are the primary identifiers for Metabolite nodes in the KG.

ChEA 2022 ChIP Enrichment Analysis 2022. Curated database of TF–target relationships derived from ChIP-seq experiments across multiple organisms. Used for TF enrichment in this KG.

GO (Gene Ontology) Standardised vocabulary of biological terms. Sub-ontologies: Biological Process (BP), Molecular Function (MF), Cellular Component (CC). GO annotations in the KG are propagated via mouse orthologs.

HMDB Human Metabolome Database. Provides metabolite annotations and cross-references. Stored on Metabolite nodes as `hmdb` field.

KEGG Kyoto Encyclopedia of Genes and Genomes. Provides pathway maps, enzyme classifications, reaction definitions, and compound information. CHO-specific pathways use organism code `cge`.

KEGG Ortholog (KO) A functional ortholog group in KEGG identified by a K-number (e.g., `K00001`). CHO genes are mapped to KOs via `assigned_to` edges, which bridge to pathways, modules, enzymes, and reactions.

NCBI Gene National Center for Biotechnology Information Gene database. NCBI Entrez Gene IDs are the primary identifiers for Gene nodes in the KG.

NCIT NCI Thesaurus. Used to standardise bioprocess parameter names (e.g., `NCIT:C74799` for a measurement parameter). Stored on Parameter nodes.

UniProt Universal Protein Resource. UniProt accessions are stored on Protein nodes as `uniprot_id` for cross-database linking.